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B. E. (Sixth Semester)
EXAMINATION, April-May, 2015
(New Course)
(Branch : Mech.)
MACHINE DESIGN—II

Time : Four Hours] [*Maximum Marks : 80*

[*Minimum Pass Marks : 28*

Note : Each question has three Parts (a), (b) and (c). Part (a) is compulsory. Solve any *one* part from (b) and (c). Attempt all questions. PSG design data book and ISI data sheets are allowed. Assume suitable data wherever necessary.

. (a) What is nipping of leaf spring ? 2

(b) A railway wagon moving at a velocity of 1.5 m/s is brought to rest by a bumper consisting of two helical springs arranged in parallel. The mass of the wagon is 1500 kg. The springs are compressed by 150 mm in bringing the wagon to rest. The spring index can be taken as 6. The springs are made of oil-hardened and tempered steel wire with ultimate tensile strength of

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1250 N/mm^2 and modulus of rigidity of 81370 N/mm^2. The permissible shear stress for the spring wire can be taken as 50% of the ultimate tensile strength. Design the spring. 14

(c) A semi-elliptic leaf spring used for the suspension of the rare axle of a truck. It consists of two extra full-length leaves and ten graduated length leaves including the master leaf. The centre to centre distance between the spring eyes in 1.2 m. The leaves are made of steel 555 *i* 2 M_0 90 (s_{yt} = 1500 N/mm^2 and E = 207000 N/mm^2) and the factor of safety is 2.5. The spring is to be designed for a maximum force of 30 kN. The leaves are prestressed so as to equalize stresses in all leaves. Determine : 14

(i) The cross-section of leaves.

(ii) The deflection at the end of the spring.

2. (a) Define beam strength of gear tooth. 2

(b) It is required to design a pair of spur gear with 20° full-depth involute teeth based on Lewis equation. The velocity factor is to be used to account for dynamic load. The pinion shaft is connected to a 10 kW, 1440 r. p. m. motor. The starting torque of the motor is 150% of the rated torque. The speed reduction is 4 : 1. The pinion as well as gear is made of plain carbon steel 40 C8 (S_{ut} = 600 N/mm^2). The factor of safety can be taken as 1.5. Design the gears, specify their

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dimensions and suggest suitable hardness for the gears. 14

(c) Design a pair of spur gears on the basis of the following data : 14

(i) Power to be transmitted = 20 kW

(ii) Pinion speed = 1400 r. p. m.

(iii) Gear ratio = 4

(iv) Material for pinion = 15Ni2Cr1Mo15 (steel)

(v) Material for gear = C-45 (steel).

3. (a) What are the advantages of helical gears over spur gears ? 2

(b) A pair of parallel helical gears consists of a 20 teeth pinion meshing with a 100 teeth gear. The pinion rotates at 720 r. p. m. The normal pressure angle is 20°, while the helix angle is 25°. The face width is 40 mm and the normal module is 4 mm. The pinion as well as gear is made of steel 40C8 (S_{ut} = 600 N/mm^2) and heat treated to a surface hardness of 300 BHN. The service factor and the factor of safety are 1.5 and 2 respectively. Assume that the velocity factor accounts for the dynamic load and calculate the power transmitting capacity of gears. 14

(c) Derive an expression for Beam strength of Bevel gear *i. e* : 14

$$S_b = m_b \sigma_b Y \left(1 - \frac{b}{A_0}\right)$$

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S_b = beam strength of tooth

m = module at the large end of tooth

b = face width

Y = Lewis form factor based on formative number of tooth

σ_b = permissible bending stress

A_0 = cone distance

4. (a) What do you mean by hydrodynamic lubrications ? 2

(b) A single-row deep groove ball bearing is subjected to a 30 seconds work cycle that consists of the following two parts : 14

	Part I	Part II
Duration(s)	10	20
Radial load (kN)	45	15
Axial load (kN)	12.5	6.25
Speed (r. p. m.)	720	1440

The static and dynamic load capacities of ball bearing are 50 and 68 kN respectively. Calculate the expected life of the bearing in hours.

(c) The following data is given for a 360° hydrodynamic bearing : 14

(i) Journal diameter = 100 mm

(ii) Bearing length = 100 mm

(iii) Radial load = 50 kN

(iv) Journal speed = 1440 r. p. m.

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(v) Radial clearance = 0.12 mm

(vi) Viscosity of lubricant = 16 Cp

Calculate :

(i) Minimum film thickness

(ii) Coefficient of friction

(iii) Power lost in friction

5. (a) Explain polygonal effect in Roller chain. 2

(b) Design a chain drive to actuate a compressor from a 10 kW electric motor at 960 r. p. m. The compressor speed is to be 350 r. p. m. Minimum centre distance should be 0.5 m. Motor is mounted on an auxiliary bed compressor is to work for 8 hrs/day. 14

(c) It is required to select a flat-belt drive for a compressor running at 720 r. p. m., which is driven by a 25 kW, 1440 r. p. m. motor. Space is available for a centre distance of 3 m. The belt is open type. 14

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